

NumPy-Only Intermediate Student Tasks

Task 1 — Missing Value Handling Using NumPy

Using only NumPy:

- Detect all NaN values
- Count missing values per column
- Replace missing values using column mean

Purpose:

Understand preprocessing without using Pandas or Scikit-learn.

Task 2 — Outlier Detection Using Z-Score

Using only NumPy:

- Calculate mean
- Calculate standard deviation
- Find rows using Z-score method

Formula:

$$z = (x - \text{mean}) / \text{standard deviation}$$

Questions:

- Which rows are outliers?
- Which feature contains most abnormal values?

Purpose:

Learn anomaly detection mathematically.

Task 3 — Feature Normalization

Normalize all numeric features using NumPy.

Formula:

$$x' = (x - x_{\min}) / (x_{\max} - x_{\min})$$

Questions:

- Which features had largest scale difference?
- How does normalization change data range?

Purpose:
Prepare data for machine learning algorithms.

Task 4 — Correlation Matrix Using NumPy

Using only NumPy:

- Generate correlation matrix
- Identify highly related features
- Identify negatively correlated features

Questions:

- Which features move together?
- Which features behave oppositely?

Purpose:
Understand feature relationships and dependency.

Task 5 — Matrix-Based Prediction System

Using only NumPy:

- Convert features into matrix form
- Create weight matrix
- Perform matrix multiplication
- Generate prediction score

Formula:
 $\text{Prediction} = XW$

Purpose:
Understand how machine learning internally uses linear algebra.